

Paper Writing

Writing a Deep Learning Paper

A written report describing original research results

The purpose: to communicate new scientific findings, clearly.

The paper includes:

- A statement about the problem or the question
- A presentation of evidence
- An assessment of the strengths/weaknesses of the presented evidence
- Conclusions

Begin by conducting a literature search

- Search on Google Scholar and Papers With Code for readings that are relevant to your interests
- Use keywords to search for related works on the topic
- Decide what topic you want to pursue for the project by reading about other experiments
- Generate ideas about how you will contribute to existing research on your own and/or with your teammates, after searching/reading other papers

Standard format

- Title
- Authors
- Abstract
- Introduction
- Related Work (“Literature Review ”)
- Methods
(Proposed Approach + Baseline)
- Results (Experimental Results)
- Discussion (Analyses)
- Conclusions and Future Work
- References

Title (and authors)

- Use the fewest possible words to describe the contents of the paper
- Consider the key words used for searching Google Scholar
- The title should capture the gist of the experiments and findings
- Never use abbreviations in titles
- Authors and their order
 - The person who writes the most gets their name listed first
 - (Establish paper writing responsibilities early)

Section 0: Abstract

It's a summary of the information in your final paper.

- Less technical
 - States the objective or scope of the experiment
 - Describes the methods
 - Summarizes the results
 - States main conclusions
 - Does not have citations
 - One paragraph, 250 words or fewer
- (Usually written last, after the paper is finished)

Section 1: Introduction

An introduction presents the nature and scope of the problem you're investigating

- Make a clear problem statement with the data and the objective explicitly defined
- State your hypothesis and propose a solution to the problem
- To facilitate understanding, you should have a high-level diagram
 - To better articulate the setup, a separate inference and training diagram may be necessary
- Terminology and basic methods should be defined at a high level

Section 2: Related Work (“Literature Review”)

In most cases, your problem will not be new; it will have been considered before and may have had a different motivation or application.

- What were the earlier solutions to your problem?
- How well did they perform in the context of your objective?
- How is your way of solving the problem different?
 - Did you apply a state-of-the-art model from one domain to a different domain?
 - Did you use a new loss to improve on an existing state-of-the-art result?
 - Did you create a new dataset and methodology that solves a novel problem?
- Important: include key ideas behind the papers cited in the literature review
- End the literature review by distinguishing yourself from the prior work

Section 3: Proposed Approach (Methodology)

To help you write this portion, we have divided the writing into three sections:

- Section 3.a: “High-Level” Concept
 - What empirical evidence motivates your hypothesis?
 - Why do you think your solution will work?
 - What makes your problem challenging and important to solve?
- Section 3.b: Technical details
 - Create a detailed diagram with core components labeled, then describe your methodology.
 - Formalize the mathematical definitions necessary to make your methodology work.
- Section 3.c: Implementation details
 - Explain what your algorithms are doing with explicit definitions
 - Outline any approximations and shortcuts you had to take
 - Provide enough detail for another researcher to replicate the study

Section 3: Proposed Approach (Baseline)

You must contextualize your proposed solution about a baseline.

- The baseline is an existing solution to your problem
- It is a competitive solution, but it has room for improvement
- The entire goal of your paper is to demonstrate that
 - The baseline you chose is valid
 - That you created a novel and a working solution to your problem
 - How your solution perform relative to the baseline
- Justify why comparison of your approach to the baseline is necessary

Section 4: Experimental Results

This section contains only facts, no analyses. State clearly what you are showing.

- Explain your data in a way such that the reader knows why you are using the data and **why** they are appropriate for your work
- Explain the “high-level” understanding of your results
- Explain **why** you got these results, including what you varied with the architecture/hyperparameters to show **how** it did/didn't work

Section 5: Analyses

- Analyze your experiment outcomes and relate them to the original hypothesis
 - Answer the research question and hypothesis you presented in the introduction
 - Discuss if your original hypothesis has been validated or invalidated.
 - Point out the strengths and limitations of the study
 - Compare and contrast your results with prior works, contextualized with your motivation
- Show the significance of the results you observed in the experiment
 - Present the relationships/generalizations that are shown by the results
 - Discuss the theoretical implications and/or practical applications of the results

Section 6: Conclusion and Future Work

- Summarize the evidence
 - What did you conclude overall in your analyses?
 - Did you solve the problem? If not, what is the way forward?
 - What future works or developments are needed to improve?

Some Rules ...

If it's hard to read, you'll get a bad grade

- Use the present tense when referring to work that has already been published
 - Example: “Deep learning is...”
- Use the past tense when referring to your study
 - Example: “We conducted...”

Some Rules ...

If it's hard to read, you'll get a bad grade

- If the method has been previously published, you should cite the reference
 - **Good:** “We used Demucs, which is a state-of-the-art music source separation model, currently capable of separating drums, bass, and vocals from the rest of the accompaniment.. Demucs is based on a U-Net convolutional architecture inspired by Wave-U-Net. The v4 version features Hybrid Transformer Demucs, a hybrid spectrogram/waveform separation model using Transformers. It is based on Hybrid Demucs (also provided in this repo) with the innermost layers are replaced by a cross-domain Transformer Encoder.” (A Defossez, et al., 2019)
 - **Not good:** “We used Demucs from GitHub for source separation of different types of audio.”

Examples of Good Papers with similar structure

- Sarenne Carrol Wallbridge, Catherine Lai and Peter Bell “Investigating perception of spoken dialogue acceptability through surprisal”
- Hansheng Chen, Pichao Wang, Fan Wang, Wei Tian, Lu Xiong, Hao Li “EPro-PnP: Generalized End-to-End Probabilistic Perspective-n-Points for Monocular Object Pose Estimation”
- Shengjia Zhao, Abhishek Sinha, Yutong (Kelly) He, Aidan Perreault, Jiaming Song, Stefano Ermon “Comparing Distributions by Measuring Differences that Affect Decision Making”

Questions During Grading (Peer Review)

- Do the authors make clear their applied or theoretical problem?
- Do the authors review and explain enough papers to clearly contextualize their work?
- Is the core model / theory / algorithm clear?
- Is the process of evaluation clear? Is it clear what the evaluation metrics are and why they are used?
- Is the process of training clear such that you could reproduce it given their model / theory / algorithm?
- Do the authors do a sufficient job experimenting with different specifications?
- Are the results presented with a clear interpretation? Are the implications of these findings clear?
- Is it clear what progress they made or what future work needs to be done?
- How close are the authors on providing an answer to their problem statement?

Thank You For Your Attention!